Muhammad Shams Ghani

Machine Learning Engineer · Computer Vision & Representation Learning

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Status: Valid UK Visa until January 2028 · Available for immediate relocation to the UK (No immediate sponsorship required)

EDUCATION

MSc Artificial Intelligence — Queen's University Belfast, UK — Sep 2024 – Sep 2025

- **MSc Dissertation:** "*Deep FOCUS: Focal Loss Optimisation for Single-Molecule Localisation Microscopy (SMLM) under Severe Sparsity*" (Supervised by Dr. Richard Gault).
- Systematically evaluated four loss functions (L1+L2, MSE, BCE, Focal Loss) across 10,000 synthetic microscopy samples generated with ImageJ ThunderSTORM.
- **Key Finding:** Demonstrated Focal Loss achieved superior reconstruction (33.8 dB PSNR, +13.6 dB vs L1+L2 and +16.7 dB vs MSE baselines) at ~20 nm spatial resolution, establishing that loss landscape geometry matters at least as much as architecture under extreme spatial class imbalance (>99.5% background).
- **Core Modules:** Deep Learning, Computer Vision, AI for Health, Machine Learning, Knowledge Engineering.

BSc Computer Science — GIK Institute of Engineering Sciences & Technology — Aug 2019 – Jun 2023

- Specialized in Artificial Intelligence, Deep Learning, Data Science, and Information Retrieval.
- **BSc Thesis:** *Text-Conditional Latent Audio Diffusion Platform* with emotion-preserving representation synthesis.

PROFESSIONAL EXPERIENCE

Graduate Trainee (AI & ML Engineering) — Itanz — Oct 2026 – Present

- **Document AI & Clinical Extraction (001):** Built pipeline rendering clinical PDFs (PyMuPDF) to local Vision-Language Models (Qwen3.5) with Pydantic validation and LLM-as-a-judge scoring.
- **Grounded RAG & Abstention (002):** Ingested 57,312 corpus documents, establishing sub-3ms exact dense retrieval in JAX with attribution and 100% precision abstention on unanswerable queries.
- **Reliability & Threat Modeling (003):** Audited RAG claims with bit-exact reproduction, enforcing 8 production quality gates, circuit breaker failovers, and OWASP Top 10 for LLMs defense.
- **Contrastive Representation Learning (004):** Fine-tuned bi-encoders via MNRL on 57,312 chunks to eliminate entity polysemy, achieving 100% Hit Rate@1 on adversarial test sets with 64s re-indexing.
- **MLOps Validation & Hybrid Serving (005):** Deployed containerized FastAPI service with Hybrid Dense + BM25 RRF (67.4% Hit@1 full-corpus), Prometheus metrics, and 0.94ms atomic model rollback.
- **Predictive ML Systems (006):** Built supervised classification pipeline on CERN Higgs benchmark across 28 continuous kinematic predictors with leakage-free stratified validation.

Machine Learning Engineer — Profundus (Computer Vision Startup, Islamabad) — Jun 2023 – Jun 2024

- Owned end-to-end ML pipeline for an eyewear virtual try-on system from dataset collection to real-time client deployment.
- Engineered neural background segmentation using contrast maps and image segmentation, eliminating manual product tagging.
- Built multimodal feature extraction and automated cataloging systems processing 10,000+ product images.

AI Engineering Intern — Profundus — Jul 2022 – Oct 2022

- Built visual search microservice for Gulzari Optics (10,000+ items) using VGG16 embeddings + Annoy via DeepImageSearch, served via FastAPI (later updated to FAISS).
- Deployed microservice with full interactive documentation, robust error handling, and performance monitoring.

OPEN SOURCE & TECHNICAL RESEARCH

OpenMouse (@openmouse/protocol) — Core Contributor & Protocol Reverse Engineer — 2026

- Reverse-engineered proprietary USB interrupt transfer reports for gaming mice via Wireshark. Authored TypeScript driver specifications and native WebHID loopback bridge routines, enabling driverless mouse configuration directly in-browser.

SCS-1 Reasoning Harness — Experimental Research Prototype 2026

- Prototyping an external executive supervisory harness for iterative LLM reasoning that tracks semantic drift between reasoning steps, checks directional NLI entailment to decide when to halt, and applies a temperature ramp when the model enters rumination loops.

LEADERSHIP & EXTRACURRICULAR

Director of External Operations & Senior Consultant — GIK Consulting Group Feb 2021 – Feb 2023

- Resolved complex engineering case study for Y Combinator-backed startup Byte (pro bono). Designed analytical training frameworks that improved junior consultant performance metrics by 50%.

President — Microsoft Learn Student Ambassadors Club 2021 – 2023

- Led technical training workshops on cloud computing and Python for 100+ students at GIKI.

TECHNICAL CAPABILITIES

Languages:	Python (Expert), TypeScript, JavaScript, C++, SQL
Deep Learning & Vision:	PyTorch (CUDA), U-Net, Diffusion Models, Vision-Language Models (VLM), OpenCV, scikit-image, TensorFlow
Retrieval & NLP:	Contrastive Learning (MNRL), SentenceTransformers, FAISS, PEFT / LoRA (Qwen), RAG Evaluation, Hugging Face
Systems & Tools:	Docker, FastAPI, WebHID API, Linux / Bash, Git, ImageJ / ThunderSTORM, LaTeX